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Big Data-Based Social Media Sentiment Analytics Platform

CSA1522- CLOUD COMPUTING AND BIG DATA ANALYTICS

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INTRODUCTION

- Social media platforms generate large volumes of unstructured data through posts, comments, hashtags, and other user-generated content.
- Analyzing such large-scale data manually is time-consuming and inefficient.
- Big Data technologies enable efficient processing of large volumes of social media data.
- Natural Language Processing (NLP) is used to analyze textual content and identify user sentiment.
- The system classifies social media content into Positive, Negative, and Neutral categories.
- The analyzed results are presented through a cloud-based visualization dashboard to identify sentiment patterns and trends.

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PROBLEM STATEMENT

- Social media platforms generate massive volumes of unstructured data every day.
- Manually analyzing large-scale social media data is time-consuming and inefficient.
- Organizations need to identify public opinions and emerging trends quickly.
- Traditional systems may face challenges in storing and processing large datasets efficiently.
- A scalable solution is required to automate sentiment analysis and extract meaningful insights from social media data.

OBJECTIVES

- To collect large volumes of social media data efficiently.
- To preprocess and clean unstructured social media text.
- To perform sentiment classification using Natural Language Processing (NLP).
- To utilize Apache Spark and Big Data technologies for efficient large-scale data processing.
- To identify sentiment patterns and trends from analyzed data.
- To present meaningful insights through an interactive cloud-based visualization dashboard.
- To develop a scalable end-to-end sentiment analytics platform.

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LITERATURE REVIEW

Paper / Research	Year	Research Focus	Approach	Limitation	Gap Addressed by Our Project
Role of Large-Scale Data Analytics Risk Information Systems in Addressing Social Disruptions in Infrastructure Development	2026	Big-data analytics for social disruptions	Large-scale data analytics	Infrastructure-specific focus	Applies analytics to general social-media sentiment
AI Sentiment Analysis-Based Brand Reputation Management Strategy in Responding to Social and Ethical Issues on Social Media	2026	Brand reputation and social-media sentiment	AI-based sentiment analysis	Focused on brand reputation	Provides broader sentiment analytics across topics
Scalable Vortex Search-Tuned Intelligent Adaptive Boosting for Sentiment Analysis of Social Media Data Using Natural Language Processing	2025	Scalable social-media sentiment analysis	NLP + Adaptive Boosting	Mainly focuses on model optimization	Combines NLP with big-data processing and visualization
Deep Learning Approaches to Sentiment Analysis and Text Classification in Social Media Data	2025	Sentiment and text classification	Deep learning + NLP	Computationally intensive	Provides a scalable end-to-end analytics platform
Integrating Sentiment Analysis and Reinforcement Learning for Equitable Disaster Response: A Novel Approach	2025	Sentiment analysis for disaster response	Sentiment analysis + reinforcement learning	Limited to disaster-response scenarios	Supports general-purpose social-media sentiment analysis

METHODOLOGY

Data Collection

Collects posts, comments , hastags and relevant data

→

Preprocessing

clean and normalise the collected data

→

Big data processing

process large scale dataset using apache spark

→

Sentiment Analysis

Apply NLP techniques to classify text as Positive, Negative, Neutral

→

Data Aggregation

Calculate Sentiment counts, percentage, trends, and keywords

→

Visualisation

Present the processed insights

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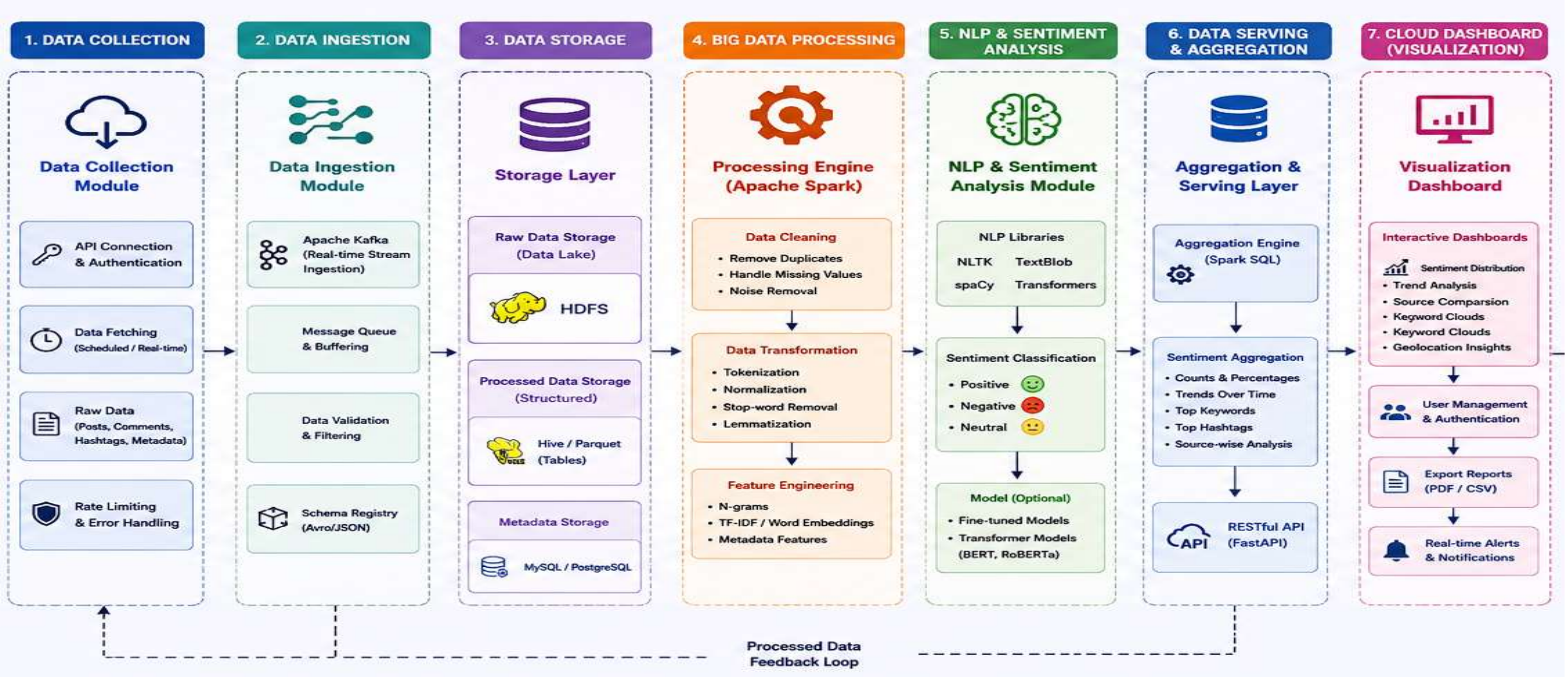
SYSTEM ARCHITECTURE



PROJECT MODULES

MODULE 1: Social Media Data Collection Module

MODULE 2: Sentiment Analysis & Big Data Processing Engine

MODULE 3: Cloud-based Visualization Dashboard

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MODULE 1: SOCIAL MEDIA DATA COLLECTION MODULE

Purpose

- ❖ Collect social-media posts, comments, hashtags, timestamps, and relevant metadata.
- ❖ Organize the collected data for further analysis.
- ❖ Prepare the data for Big Data processing and sentiment analysis.

Process

- ❖ Data collection from available social-media sources/APIs.
- ❖ Data validation and basic cleaning.
- ❖ Storage of collected data in a structured format.

Input

Social Media Data

Output

Structured Dataset

MODULE 1 IMPLEMENTATION

	post_id	platform	timestamp	username	text	language	topic	sentiment	sentiment_	emotion	likes	shares	comments	hashtags	location	engagement_rate
2	SM_0001	Instagram	2026-01-0	user_7623	Very disap	English	E-commerc	Negative	-0.664	sadness	35741	356	1728	#Shopping	Online	100
3	SM_0002	YouTube	2026-03-2	user_5411	So happy w	English	Education	Positive	0.865	excitement	49729	1307	2859	#Students	Bengaluru	100
4	SM_0003	Facebook	2026-05-2	user_8567	This Gamin	English	Gaming	Positive	0.708	optimism	35142	511	1550	#Gaming	Delhi	100
5	SM_0004	Facebook	2026-02-2	user_8984	இந்த upd	Tamil	Entertainm	Negative	-0.699	anger	23909	666	758	#Music #Er	Mumbai	55.46
6	SM_0005	Reddit	2026-02-2	user_6258	This Food l	Hinglish	Food	Negative	-0.918	frustration	14392	1402	332	#Restauran	Hyderabad	100
7	SM_0006	Facebook	2026-02-0	user_8662	Feeling rea	English	E-commerc	Positive	0.936	optimism	42129	3758	585	#Deals	Kolkata	100
8	SM_0007	Instagram	2026-04-1	user_6001	A new repc	English	Finance	Neutral	-0.041	calm	10016	642	163	#Finance	Online	12.8
9	SM_0008	YouTube	2026-03-2	user_3073	The latest T	English	Technology	Neutral	-0.075	calm	35190	3075	1092	#Tech	Coimbator	50.56
10	SM_0009	Instagram	2026-04-0	user_8850	Excellent q	Hinglish	Finance	Positive	0.978	joy	33270	2494	814	#Investing	Pune	100
11	SM_0010	Facebook	2026-07-1	user_8206	The E-com	English	E-commerc	Negative	-0.581	disappoint	5613	497	35	#OnlineShc	Mumbai	9.4
12	SM_0011	Instagram	2026-04-2	user_1839	இந்த ser	Tamil	Finance	Positive	0.736	excitement	42590	3059	1794	#Investing	Pune	74.67
13	SM_0012	Instagram	2026-03-0	user_3808	Great job!	English	Technology	Positive	0.651	satisfaction	2058	84	18	#Innovatio	Madurai	3.29
14	SM_0013	Instagram	2026-08-1	user_1794	இந்த ser	Tamil	Healthcare	Positive	0.804	joy	23219	1734	841	#Health #H	Kolkata	44.92
15	SM_0014	Instagram	2026-01-2	user_2222	The Financ	English	Finance	Negative	-0.881	anger	16371	154	907	#Money #N	Delhi	24.67
16	SM_0015	Reddit	2026-04-1	user_3028	This is exac	English	Technology	Positive	0.762	satisfaction	17380	1605	931	#Tech	Delhi	57.27
17	SM_0016	YouTube	2026-05-1	user_1898	Absolutely	English	Technology	Positive	0.805	excitement	32954	2175	644	#Tech #AI	Online	100
18	SM_0017	YouTube	2026-05-1	user_3677	So happy w	English	E-commerc	Positive	0.728	optimism	27474	2692	1195	#Shopping	Pune	46.15
19	SM_0018	Facebook	2026-02-2	user_5574	What a wo	English	Healthcare	Positive	0.608	excitement	36896	409	300	#Health #V	Bengaluru	53.4
20	SM_0019	Reddit	2026-03-1	user_4466	Several use	English	Finance	Neutral	-0.065	curiosity	42858	64	2735	#Investing	Bengaluru	63.9
21	SM_0020	Facebook	2026-08-0	user_4626	The latest (	English	Gaming	Neutral	-0.103	calm	17300	1035	1000	#Gamer #E	Chennai	61.54
22	SM_0021	Facebook	2026-08-1	user_8151	Best decisi	English	Sports	Positive	0.83	excitement	36759	39	458	#Football	Chennai	100

CSV import

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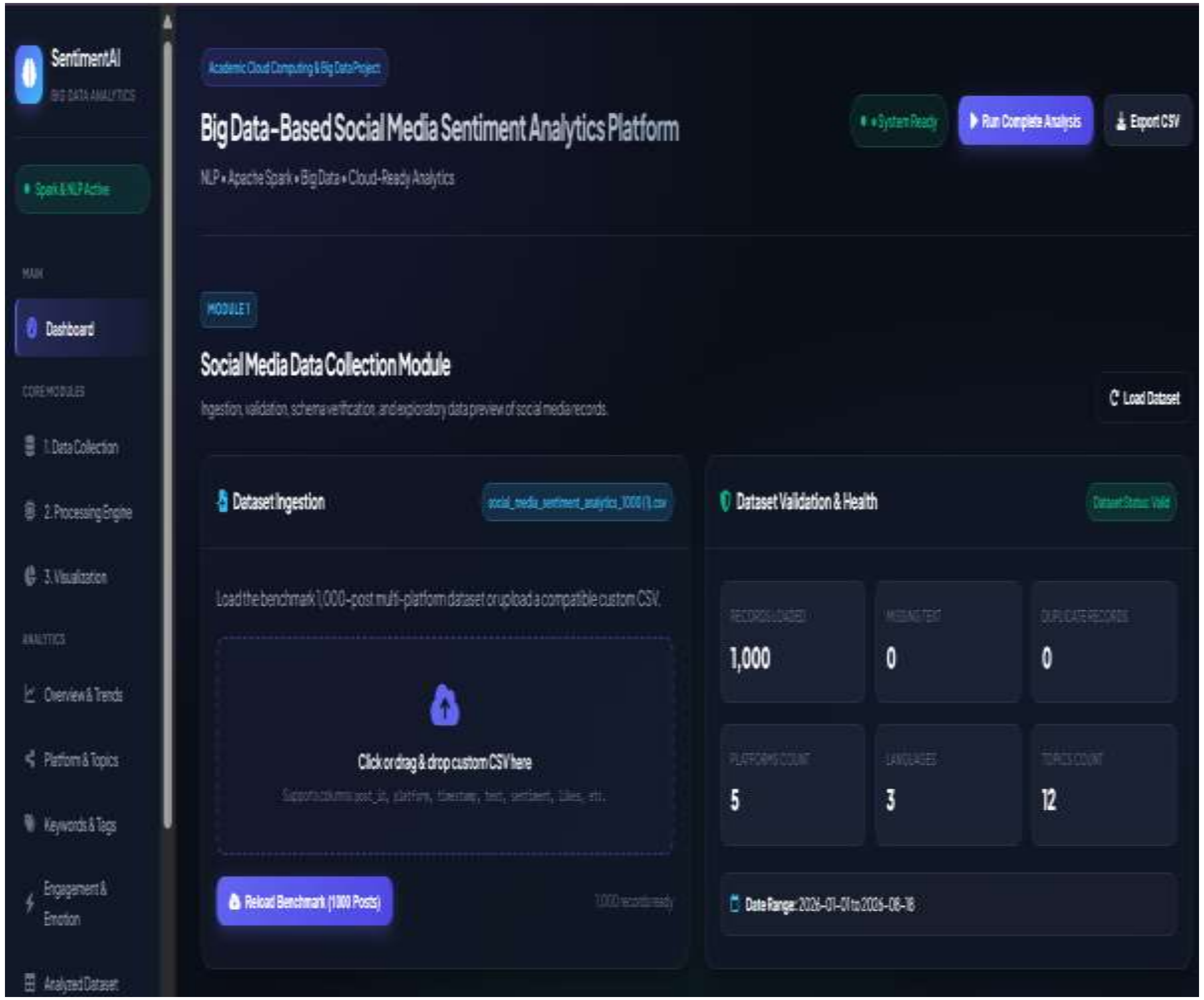
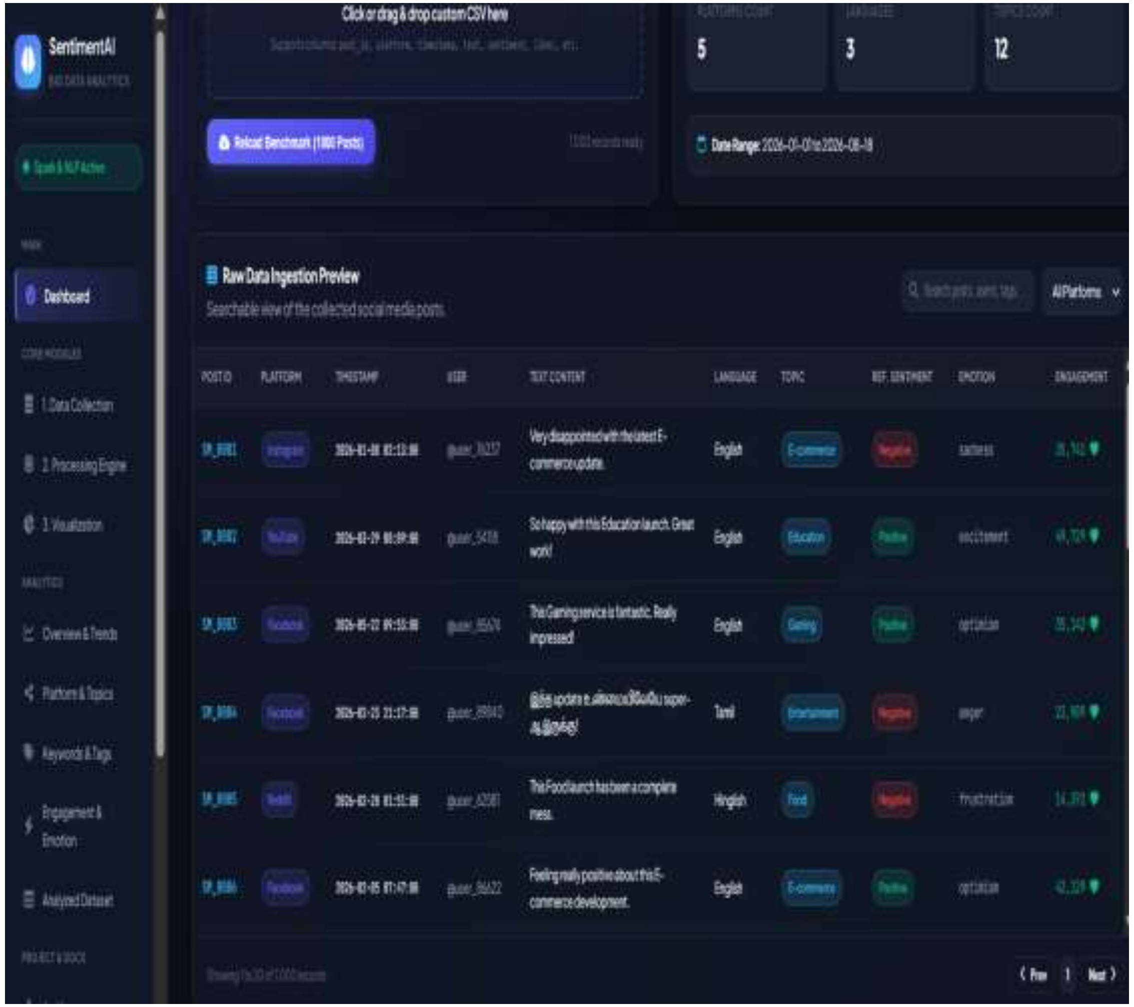




MODULE 2: SENTIMENT ANALYSIS & BIG DATA PROCESSING ENGINE

Purpose:

- Process large volumes of collected social-media text efficiently.
- Prepare the data for automated sentiment analysis.
- Identify the sentiment expressed in each social-media post.

Process:

- Clean and preprocess the collected text.
- Perform tokenization and stop-word removal.
- Use Apache Spark for large-scale data processing.
- Apply NLP-based sentiment analysis.
- Classify posts into Positive, Negative, or Neutral categories.

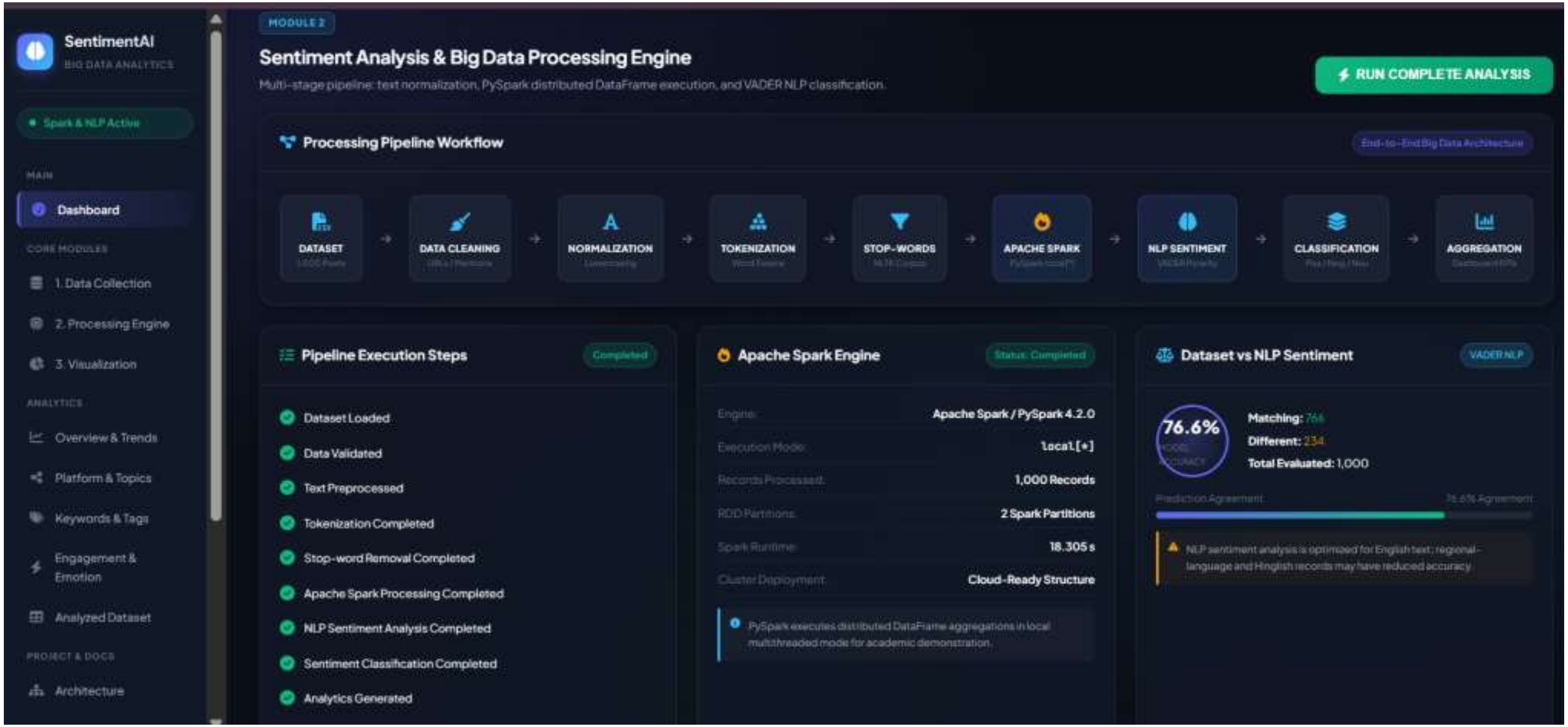
Input

Collected & Structured Social Media Data

Output

Analyzed Sentiment Results

MODULE 2 IMPLEMENTATION



MODULE 3:CLOUD-BASED VISUALIZATION DASHBOARD

Purpose:

- Present the analyzed sentiment results through an interactive dashboard.
- Provide a clear view of sentiment patterns and trends.
- Enable easy interpretation of the analyzed social-media data.

Process:

- Receive analyzed sentiment results from Module 2.
- Aggregate sentiment data into meaningful metrics.
- Generate charts, graphs, and tables.
- Display sentiment distribution, trends, keywords, and hashtags.
- Provide an interactive interface for exploring the results.

Input:

Analyzed Sentiment Results

Output:

Interactive Cloud-Based Dashboard → Sentiment Insights

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MODULE 3 IMPLEMENTATION

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RESULTS AND DISCUSSION

- The proposed platform establishes an end-to-end workflow for collecting, processing, and analyzing social-media data.
- The system processes social-media text using NLP and Apache Spark and classifies posts into Positive, Negative, and Neutral sentiments.
- The integration of Big Data processing and sentiment analysis enables scalable analysis of large volumes of social-media content.
- The cloud-based dashboard presents the analyzed results through interactive charts, graphs, and tables for easier interpretation.
- The complete implementation will be further evaluated using actual datasets and performance metrics

CHALLENGES AND LIMITATIONS

- Handling large volumes of unstructured social-media data can increase processing and storage requirements.
- Noisy text, slang, abbreviations, and informal language can affect sentiment classification.
- Sarcasm and context-dependent expressions are difficult for sentiment models to interpret accurately.
- Social-media API availability and access restrictions can affect continuous data collection.
- The accuracy of the analysis depends on the quality and representativeness of the dataset.
- The initial implementation may be limited to selected social-media platforms and languages.

FUTURE SCOPE

- Real-time sentiment analysis to monitor social-media opinions and detect sudden changes instantly.
- Multilingual and emotion analysis to support regional languages and identify emotions like happiness, anger, and sadness.
- AI-based prediction to predict future sentiment trends and public opinion using advanced ML models.
- Fake news and misinformation detection to identify misleading content and spam on social-media platforms.
- Advanced visualization and alerts to provide interactive dashboards, geographical insights, and real-time notifications.

CONCLUSION

- The proposed platform provides an end-to-end solution for social-media sentiment analytics.
- NLP-based sentiment analysis classifies social-media content into Positive, Negative, and Neutral categories.
- The system integrates data collection, processing, sentiment analysis, and visualization into a unified platform.
- The cloud-based dashboard presents sentiment patterns, trends, keywords, and hashtags as meaningful visual insights.
- The platform provides a scalable foundation for future real-time and advanced sentiment analytics.

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